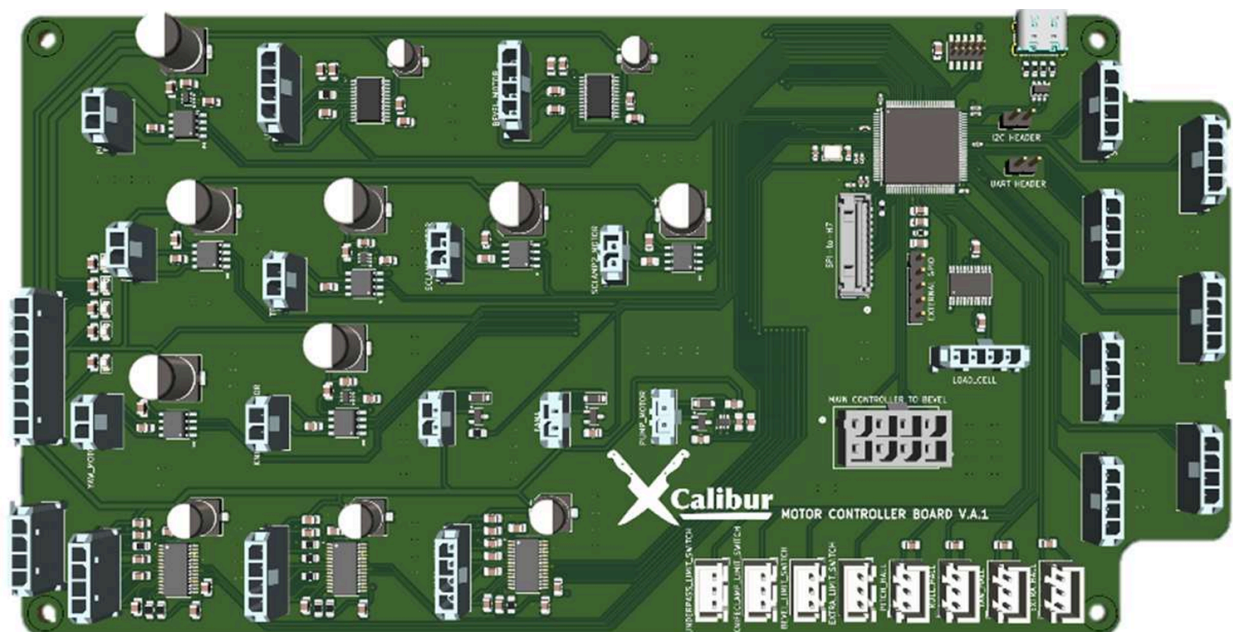
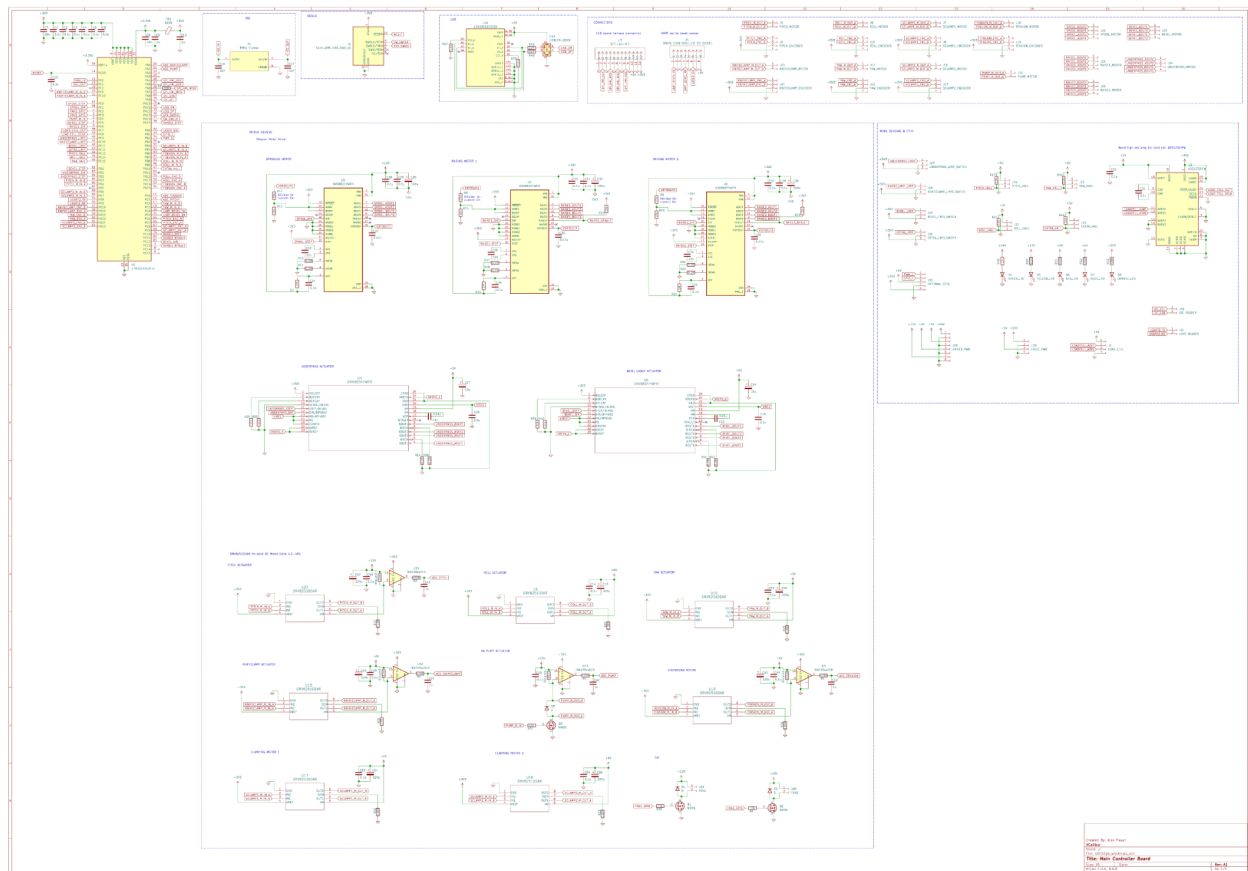


XCalibur Main Control Board Design



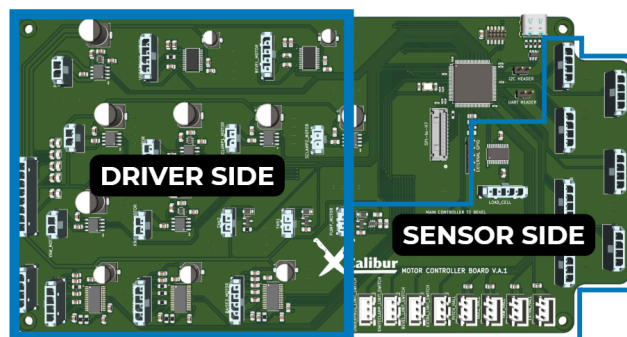
General Notes:

The STM32G491 was selected as the main controller due to its balance of processing performance, advanced analog features, and versatile peripherals. Its 170 MHz Cortex-M4 core enables reliable real-time control, while the integrated high-resolution timers, fast ADCs, comparators, and op-amps support precise motor control and sensor interfacing without additional hardware. The device also offers a wide range of communication interfaces, making it well suited to act as the central hub for coordinating multiple subsystems. The selected size is also a large package size providing many gpio, and communication bus lines which is important considering the many peripherals in this design. The snippet of the schematic for the STM32G4 can be found in the figure below.

- The purpose of this board is to support actuation and connect peripheral PCBs to a main processing unit.
- Utilizes an STM32G4 as the main processing unit, and several DRV stepper and h-bridge drivers, with some simple MOSFET drivers.
- Applied a form of power domain isolation to reduce noise from motor drivers that can impact analog or digital sensing signals.
- Integrated RC filters on analog signals to smoothen out noise that is caused by parasitic capacitance due to the need to use vias.
- Performed complete assembly and testing of the PCB.

Small cool things to notice:

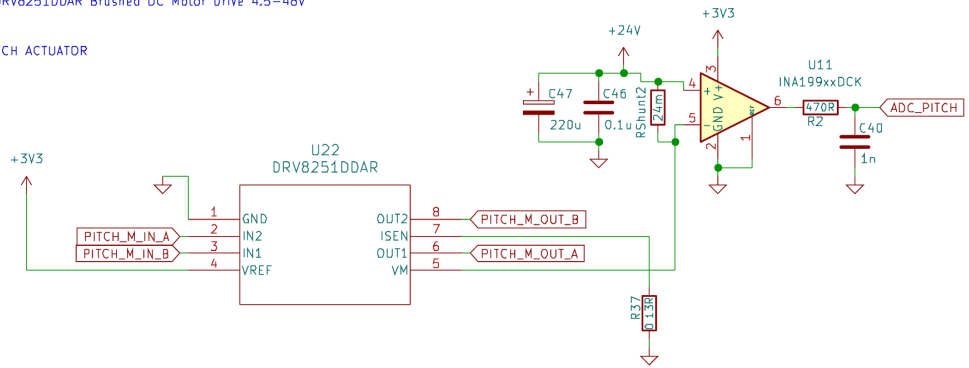
- To reduce cross talk, the driver side was separated from the sensing side. All drivers were put on one side, to also centralize the majority of the heat which can be resolved with a heatsink+fan combo onto that area.
- The arrangement of components could have been tighter, but due to tight deadlines and it being a fydp project, there was not enough time to refine the design.



- To measure the pressure applied on the knife from the sharpening system, a current sensor was added instead to measure how much torque the motor used, which reflects on the pressure. An RC filter was added on the ADC output since the trace had to use a via. This helps omit the parasitic capacitances.

DRV8251DDAR Brushed DC Motor Drive 4.5–48V

PITCH ACTUATOR



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